PEMROGRAMAN BERBASIS FRAMEWORK
JOBSHEET 4

NAMA : Muhammad Fahreza Rohmansyah
KELAS : 3E
NO. ABSEN : 15
NIM : 2341720151

PROGRAM STUDI TEKNIK INFORMATIKA
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI MALANG
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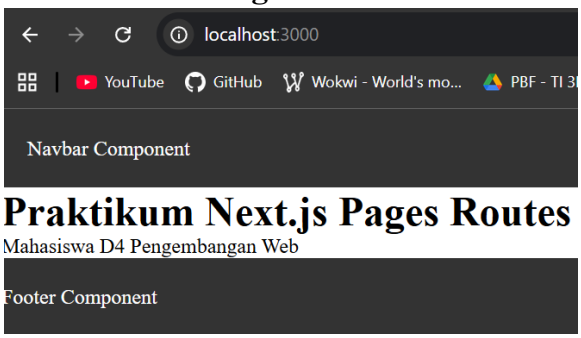
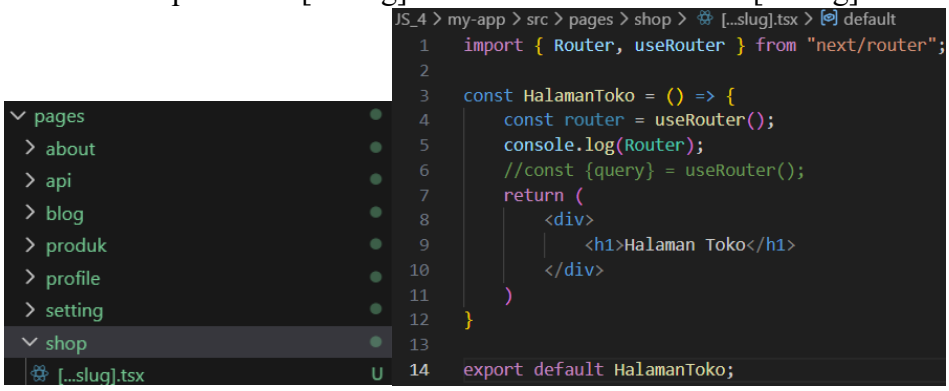
Praktikum

1. Jalankan Program

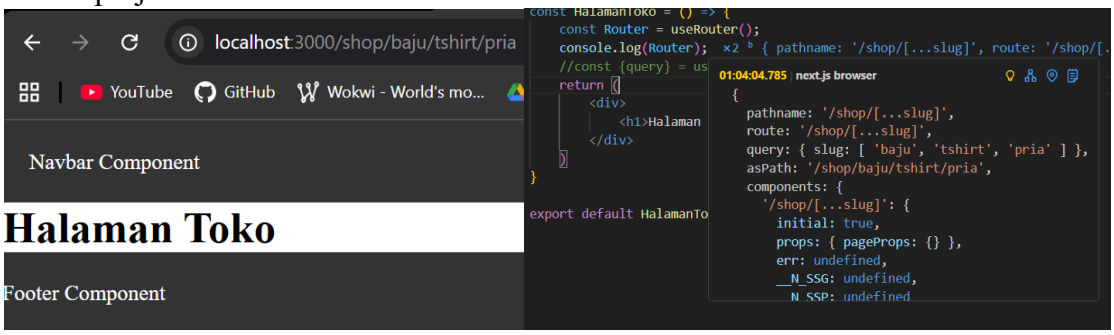


2. Membuat Catch-All Route

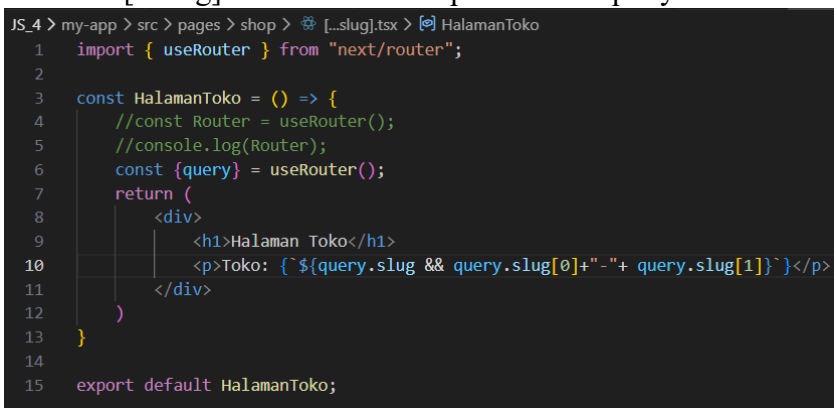
- Buat folder shop dan file […slug].tsx lalu modifikasi Isi file […slug].tsx



- Jalankan project dan cek console



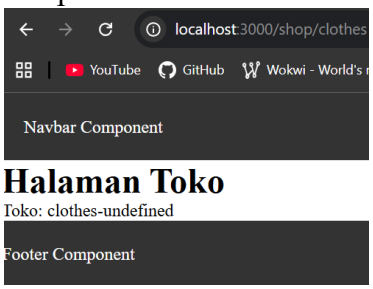
- Modifikasi [...slug].tsx untuk menampilkan nilai query



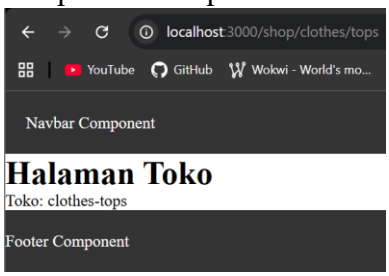
3. Pengujian Catch-All Route

- Akses URL berikut di browser:

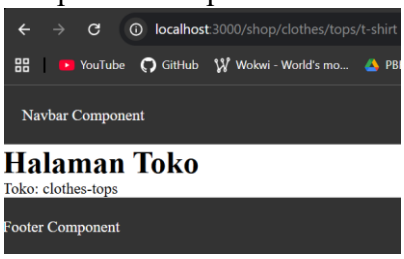
  /shop/clothes



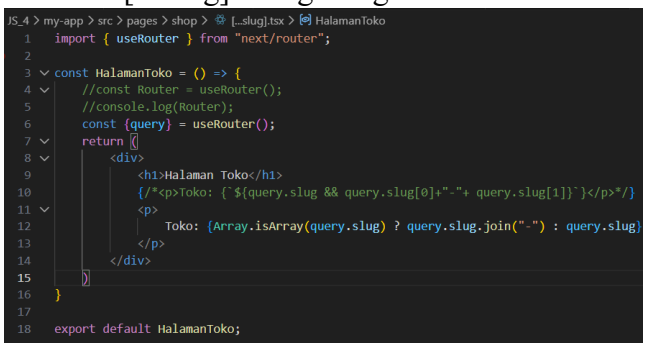
  /shop/clothes/tops



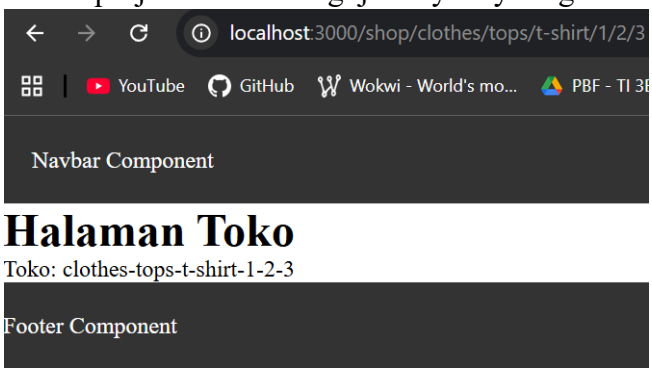
  /shop/clothes/tops/t-shirt



- Modifikasi […slug].tsx agar segmentasi tidak terbatasi

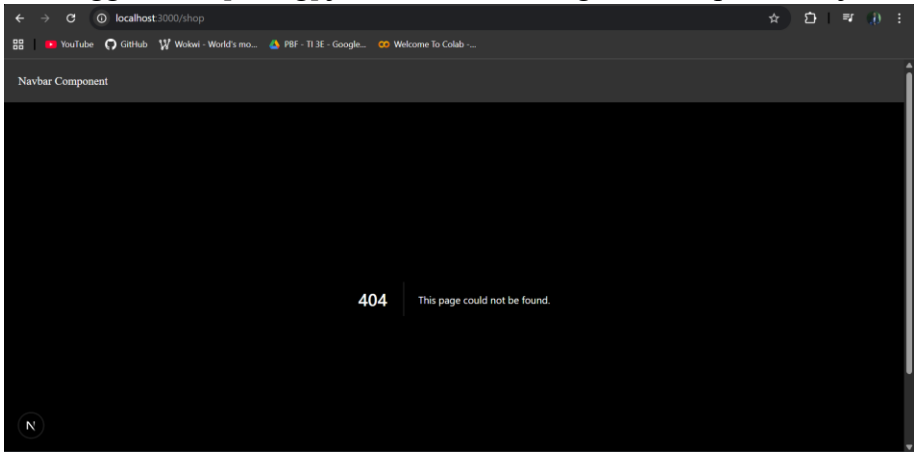


- Jalankan project untuk menguji banyaknya segmentasi yang terbaca

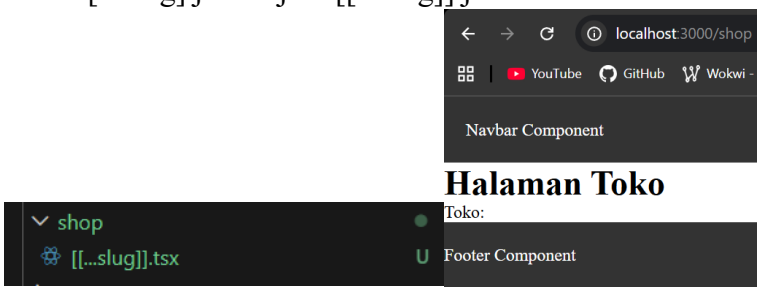


4. Optional Catch-All Route

- Jika menggunakan [...slug].js maka ketika mengakses shop akan terjadi error



- Rename [...slug].js menjadi [[...slug]].js

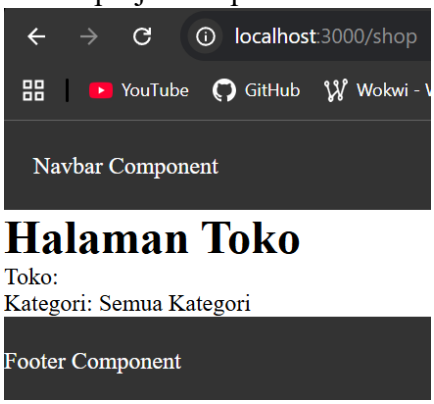


5. Validasi Parameter

- Tambahkan validasi agar tidak error saat slug kosong

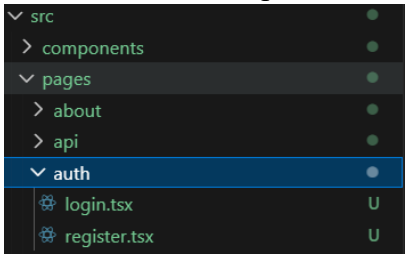
```
JS_4 > my-app > src > pages > shop > [[...slug]].tsx > default
import { useRouter } from "next/router";

const HalamanToko = () => {
    //const Router = useRouter();
    //console.log(Router);
    const {query} = useRouter();
    const slug = query.slug as string[] | undefined;
    return (
        <div>
            <h1>Halaman Toko</h1>
            {/*<p>Toko: {`${query.slug && query.slug[0]+"-"+ query.slug[1]}`}</p>*/}
            <p>
                Toko: {Array.isArray(query.slug) ? query.slug.join("-") : query.slug}
            </p>
            <p>Kategori:  {slug ? slug[0] : "Semua Kategori"}</p>
        </div>
    )
}

export default HalamanToko;
```

- Jalankan project tanpa menambahkan parameter



6. Membuat Halaman Login & Register

- Membuat folder /auth pada folder pages beserta file login.tsx dan register.tsx di dalamnya



- Modifikasi file login.tsx

```
JS_4 > my-app > src > pages > auth > login.tsx > default
import Link from "next/link";

const HalamanLogin = () => {
    return (
        <div>
            <h1>Halaman Login</h1>
            <Link href="/auth/register">Ke Halaman Register</Link>
        </div>
    );
};

export default HalamanLogin;
```

- Modifikasi file register.tsx

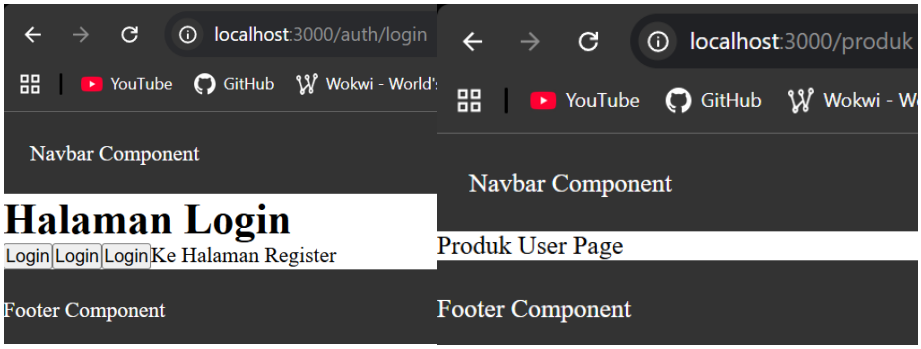
```
JS_4 > my-app > src > pages > auth > register.tsx > default
import Link from "next/link";

const HalamanRegister = () => {
    return (
        <div>
            <h1>Halaman Register</h1>
            <Link href="/auth/login">Ke Halaman Login</Link>
        </div>
    );
};

export default HalamanRegister;
```

7. Navigasi Imperatif (router.push)

- Tambahkan button login pada login.tsx

```
JS_4 > my-app > src > pages > auth > login.tsx > HalamanLogin
import Link from "next/link";
import { useRouter } from "next/router";

const HalamanLogin = () => {
    const {push} = useRouter();
    const handleLogin = () => {
        // Login logic here
        push("/produk");
    };
    return (
        <div>
            <h1>Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>
            <button onClick={() => push('/produk')}>Login</button>
            <button onClick={() => handleLogin()}>Login</button>
            <Link href="/auth/register">Ke Halaman Register</Link>
        </div>
    );
};

export default HalamanLogin;
```

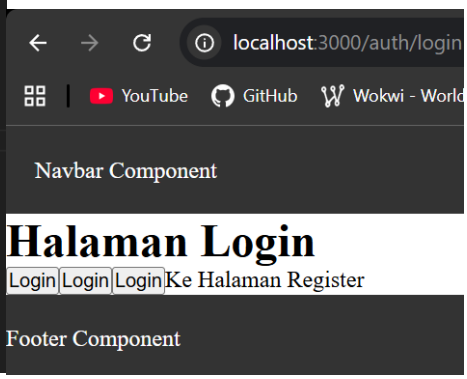
- Pengujian button login



8. Simulasi Redirect (Belum Login)

- Modifikasi file produk agar jika belum login diarahkan ke halaman login, jadi saat mengakses halaman produk nanti akan otomatis diarahkan ke halaman login

```
JS_4 > my-app > src > pages > produk > index.tsx > produk > use
import { useRouter } from "next/router";
import { useEffect,useState } from "react";

const produk = () => {
    const [isLogin, setIsLogin] = useState(false);
    const {push} = useRouter();
    useEffect(() => {
        if (!isLogin) {
            push("/auth/login");
        }
    }, []);
    return (
        <div>
            Produk User Page
        </div>
    );
};

export default produk;
```

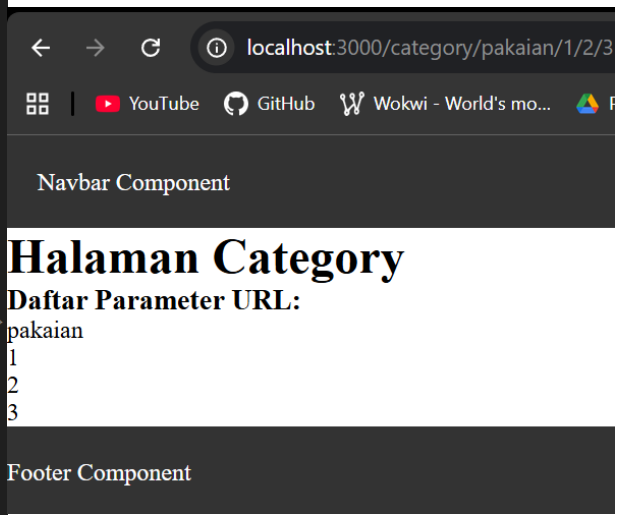


Tugas Praktikum

Tugas 1 (Wajib)

- Buat catch-all route:
- /category/[...slug].js
- Tampilkan seluruh parameter URL dalam bentuk list.

```
JS_4 > my-app > src > pages > category > [...slug].tsx > default
import { useRouter } from "next/router";

const CategoryPage = () => {
    const router = useRouter();
    const { slug } = router.query;

    return (
        <div>
            <h1>Halaman Category</h1>

            <h3>Daftar Parameter URL:</h3>

            <ul>
                {Array.isArray(slug) ? (
                    slug.map((item, index) => (
                        <li key={index}>{item}</li>
                    ))
                ) : (
                    <li>Tidak ada parameter</li>
                )}
            </ul>
        </div>
    );
};
```



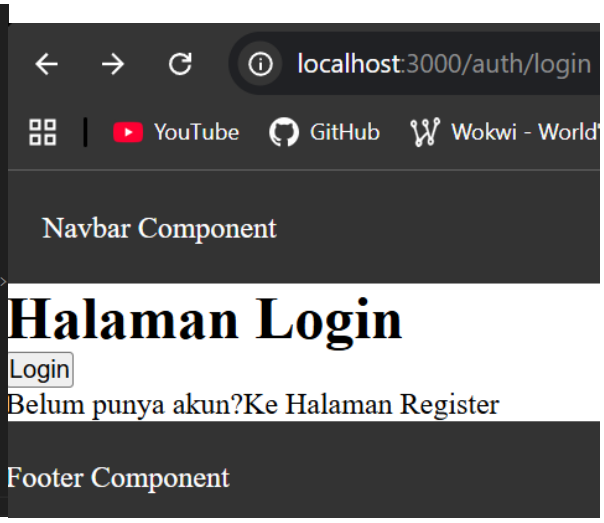
Tugas 2 (Wajib)

- Buat navigasi:
- Login → Product (imperatif)
- Login ↔ Register (Link)

➢ Halaman Login

JS_4 > my-app > src > pages > auth > login.tsx > default

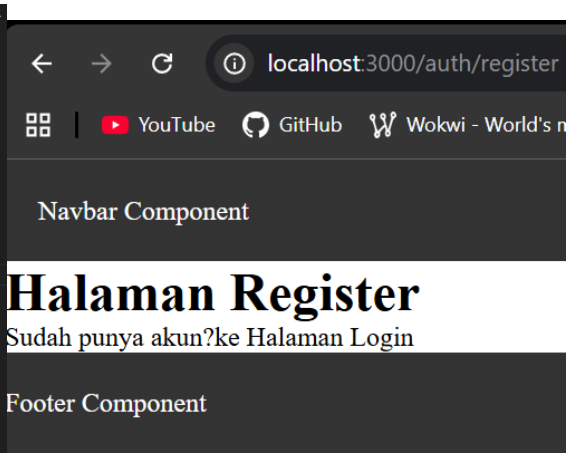
```python
import Link from "next/link";
import { useRouter } from "next/router";

const HalamanLogin = () => {
    const router = useRouter();
    const handleLogin = () => {
        // Login logic here
        router.push("/produk");
    };
    return (
        <div>
            <h1>Halaman Login</h1>
            <button onClick={handleLogin}>Login</button>
            <p>
                Belum punya akun?
                <Link href="/auth/register">
                    Ke Halaman Register
                </Link>
            </p>
        </div>
    );
};

export default HalamanLogin;
```



➢ Halaman Register

JS_4 > my-app > src > pages > auth > register.tsx > HalamanRegister

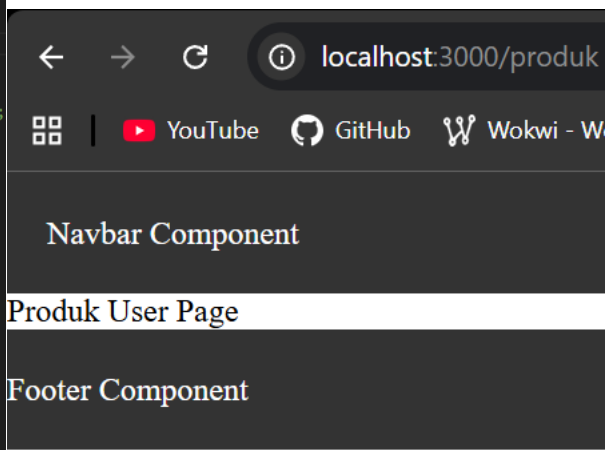
```python
import Link from "next/link";

const HalamanRegister = () => {
    return (
        <div>
            <h1>Halaman Register</h1>

            <p>
                Sudah punya akun?
                <Link href="/auth/login">
                    ke Halaman Login
                </Link>
            </p>
        </div>
    );
};

export default HalamanRegister;
```



➢ Halaman Produk

JS_4 > my-app > src > pages > produk > index.tsx > ...

```python
import { useRouter } from "next/router";
import { useEffect,useState } from "react";

const produk = () => {
    /*const [isLogin, setIsLogin] = useState(false);
    const {push} = useRouter();
    useEffect(() => {
        if (!isLogin) {
            push("/auth/login");
        }
    }, []);*/
    return (
        <div>
            Produk User Page
        </div>
    );
};

export default produk;
```



Tugas 3 (Pengayaan)

- Terapkan redirect otomatis ke login jika user belum login.

JS_4 > my-app > src > pages > user > index.tsx > default

```
import { useRouter } from "next/router";
import { useEffect,useState } from "react";

const UserSettingPage = () => {
    const [isLogin, setIsLogin] = useState(false);
    const {push} = useRouter();
    useEffect(() => {
        if (!isLogin) {
            push("/auth/login");
        }
    }, []);
    return (
        <div>
            User Setting Page
        </div>
    );
};

export default UserSettingPage;
```

Pertanyaan Refleksi

1. Apa perbedaan [id].js dan [...slug].js?
   Jawab:

- [id].js: Menangkap satu segmen URL saja (contoh: /produk/baju)
- [...slug].js: Menangkap banyak segmen sekaligus dalam satu file (contoh: /produk/baju/pria/merah)

2. Mengapa slug berbentuk array?
   Jawab:
   Karena Catch-All Route mengambil semua bagian URL yang dipisahkan garis miring (/). Agar tiap bagian tersebut bisa diolah secara terpisah oleh kode, Next.js menyimpannya dalam struktur array
3. Kapan sebaiknya menggunakan Link dan router.push()?
   Jawab:

- Link (Deklaratif): Digunakan untuk navigasi standar melalui elemen yang diklik pengguna (seperti menu atau tombol navigasi).
- router.push() (Imperatif): Digunakan untuk navigasi yang dipicu oleh logika kode, seperti *redirect* otomatis setelah login sukses.

4. Mengapa navigasi Next.js tidak me-refresh halaman?
   Jawab:
   Karena Next.js menggunakan Client-side Navigation. Hanya data/konten yang diperlukan saja yang dimuat ulang, bukan seluruh dokumen HTML-nya, sehingga transisi halaman terasa instan dan mulus